

Explain It Like You Built It: ReAct Tool Calling & Anti-Hallucination

Assignment: Week 6 (General AI Fluency) | **Author:** Muhammad Akbar Pradana (ML Intern @ FlyRank AI)
System Analyzed: Personal ML & Search Intelligence Scout Agent (agent/search_scout_agent.py & tools.py)

1. What Part of the Build I Am Explaining

When building the **Personal ML & Search Intelligence Scout Agent** in Week 5 (FL-07), the mechanism I found most critical and interesting is the **ReAct (Reasoning + Acting) Tool Calling Loop** that allows a Language Model to query private tabular datasets without hallucinating numbers.

2. The Core Problem: Why Standard Chatbots Fail in Data Science

If you prompt a standard conversational LLM (like vanilla ChatGPT) with: *'What are the 90-day search impressions and SERP rank for content_9532f197bbc8?'*, the model will usually fabricate plausible-sounding, fake numbers. Because the model has no direct connection to private company datasets, its generative objective forces it to predict the most statistically probable sequence of tokens rather than factual truth.

3. How We Solved It: The 3-Step ReAct Tool Calling Architecture

In our implementation, we decoupled natural language comprehension from factual data retrieval:

Stage	Component	Technical Operation
1. Reasoning	Intent & Entity Parser	Regex parses content ID (content_9532f197bbc8) and maps intent to decay audit.
2. Acting	Local Python Tool Dispatcher	Executes tool_audit_decay() on 30k-row CSV, retrieving 309k impr, pos 2.0, 104d stale.
3. Observing	Grounded Formatter	Returns structured JSON, assigns Reason Code page_one_decay_risk & Action [REFRESH].

4. Key Takeaways & Lessons Learned

- **Separation of Concerns:** LLMs are superior at natural language parsing and summarization, but Python functions are superior at arithmetic, indexing, and tabular filtering.
- **Human-in-the-Loop Ownership:** By writing custom tools with explicit error handling (e.g., handling avg_position == 0 missing values and Windows UTF-8 terminal encoding), we ensure the system is production-safe and fully explainable.